

Databricks & Production Data Engineering with PySpark

Advanced Practical Training | Total Duration: 12 Hours | 6 Sessions × 2 Hours

Target Audience	Experienced Data Engineers / IT Professionals (10+ years)
Prerequisites	Basic Python, SQL and data engineering concepts
Focus	Databricks, PySpark, Delta Lake, production data pipelines, optimization and real-world engineering practices

Session 1 — Databricks & Lakehouse Architecture

- Introduction to Databricks and its role in modern data engineering
- Workspace, notebooks, compute and Databricks Runtime
- Apache Spark architecture: Driver, Executors and distributed processing
- Lakehouse architecture and Medallion Architecture — Bronze, Silver and Gold
- Databricks data engineering workflow
- Working with tables, files and data sources
- Production architecture considerations

Hands-on: Create a basic Databricks data engineering workflow.

Session 2 — PySpark for Production Data Engineering

- DataFrames and schemas
- Reading CSV, JSON, Parquet and Delta data
- Transformations: select, filter, withColumn, when and aggregations
- Joins and Window Functions
- Spark SQL
- Handling nulls, duplicates and bad records
- Schema definition and schema evolution concepts
- Batch processing patterns
- Distributed processing concepts and common PySpark mistakes

Hands-on: Develop a PySpark transformation pipeline using realistic data.

Session 3 — Delta Lake & Incremental Data Processing

- Why Delta Lake? Parquet vs Delta
- Delta tables and ACID transactions
- Create, read and write Delta tables
- Append, overwrite and incremental processing
- MERGE / UPSERT patterns
- Insert, Update and Delete operations
- Schema enforcement and schema evolution
- Time Travel and table history

- SCD Type 1 and SCD Type 2
- Table maintenance and performance concepts

Hands-on: Build an incremental Delta pipeline with MERGE and SCD handling.

Session 4 — Data Ingestion, ETL/ELT & Data Quality

- Modern data ingestion architecture
- Full load vs incremental load
- Batch vs streaming concepts
- File-based and API/database ingestion patterns
- Auto Loader and incremental file ingestion
- Schema inference and schema evolution
- CDC concepts
- Data cleansing and standardization
- Data quality and validation rules
- Error handling and bad-record/quarantine patterns
- Bronze → Silver → Gold pipeline design

Hands-on: Build an end-to-end ingestion and transformation pipeline.

Session 5 — Databricks Jobs, Orchestration & Production Practices

- Databricks Jobs and task dependencies
- Scheduling and parameterization
- Retry and failure-handling strategies
- Pipeline orchestration
- Development → Test → Production practices
- Git/version-control concepts
- CI/CD concepts for data pipelines
- Logging and monitoring
- Troubleshooting production failures
- Operational best practices

Hands-on: Create and schedule a production-style Databricks Job.

Session 6 — Performance Optimization, Governance & Capstone

- Spark execution plan and lazy evaluation
- Shuffle and partitioning
- Repartition vs Coalesce
- Broadcast joins and data skew
- Caching and persistence
- Small-file problem
- Driver vs Executor issues
- Spark UI and performance troubleshooting
- Delta optimization and maintenance

- Unity Catalog fundamentals: catalogs, schemas, tables, permissions and lineage
- End-to-end production architecture review

Capstone: Source → Ingestion → Bronze → Silver → Gold → Delta → Job → Monitoring, combining the major concepts from all sessions.

Learning Outcomes

- Design and develop production-oriented data engineering pipelines using Databricks and PySpark.
- Work confidently with Delta Lake, incremental processing, MERGE and SCD patterns.
- Build Bronze/Silver/Gold Lakehouse pipelines and implement data quality controls.
- Understand Databricks Jobs, orchestration, monitoring and production troubleshooting.
- Identify and resolve common Spark performance issues such as shuffle, skew, partitioning and driver overload.
- Apply practical governance and deployment concepts to enterprise data platforms.

Training Approach: Approximately 70% hands-on and 30% concepts/architecture, with emphasis on production scenarios, troubleshooting, performance optimization and enterprise data engineering.

Note: Azure fundamentals and Python fundamentals are covered separately in the companion modules; this syllabus focuses on Databricks, PySpark and production data engineering.

Companion Modules — Python & Microsoft Azure

Added to the Databricks & Production Data Engineering with PySpark syllabus

Module A — Python for Data Engineering

Suggested Duration: 12 Hours | 6 Sessions × 2 Hours

Objective: Build strong practical Python skills required for data engineering, automation, APIs and cloud data workflows.

Session 1 — Python Fundamentals

- Python syntax, variables, data types and operators
- Strings, lists, tuples, sets and dictionaries
- Conditional statements and loops
- Functions, parameters, return values and scope
- List/dictionary comprehensions
- Modules, packages and virtual environments

Hands-on: Build a Python data-processing utility.

Session 2 — Object-Oriented Python

- Classes and objects
- Constructors, instance/class/static methods
- Inheritance, polymorphism and encapsulation
- Abstract classes and interfaces concepts
- Exception handling and custom exceptions
- Clean code and reusable components

Hands-on: Develop a reusable ETL utility using OOP.

Session 3 — Python for Data Processing

- File handling: CSV, JSON, Excel and text files
- NumPy fundamentals
- Pandas Series and DataFrames
- Filtering, grouping, merging and aggregation
- Missing values and duplicate handling
- Data transformation and validation

Hands-on: Clean and transform a real-world dataset.

Session 4 — APIs, Database & Automation

- REST API concepts and HTTP methods
- Consuming APIs with Python
- JSON request/response handling
- PostgreSQL/MySQL connectivity
- SQL execution from Python
- Environment variables and configuration management

Hands-on: Build an API-to-database ingestion script.

Session 5 — Testing, Logging & Production Python

- Logging and debugging
- Unit testing with pytest/unittest concepts
- Type hints and code quality
- Requirements and dependency management
- Git and version-control practices
- Error handling and production-ready scripts

Hands-on: Test and productionize a Python data pipeline.

Session 6 — Python Data Engineering Capstone

- End-to-end Python ETL workflow
- Extract from API/file/database
- Transform with Pandas/Python
- Load into a relational database or cloud storage
- Logging, validation and error handling

Capstone: Build a production-style Python ETL pipeline.

Module B — Microsoft Azure for Data Engineering

Suggested Duration: 12 Hours | 6 Sessions × 2 Hours

Objective: Understand Azure services used to build, deploy, secure and monitor modern cloud data engineering solutions.

Session 1 — Azure Fundamentals

- Cloud computing concepts and Azure overview
- Azure Portal, subscriptions and resource groups
- Regions, availability zones and Azure resources
- Azure Storage fundamentals
- Blob Storage and Data Lake Storage Gen2
- Identity and access fundamentals

Hands-on: Create Azure resources and configure cloud storage.

Session 2 — Azure Data Storage & Databases

- Azure Data Lake Storage Gen2 architecture
- Containers, directories and access control
- Azure SQL Database fundamentals
- Azure Cosmos DB overview
- Storage security and lifecycle management
- Data lake folder and partitioning strategies

Hands-on: Create a cloud data lake structure for an ETL project.

Session 3 — Azure Data Factory

- Azure Data Factory architecture
- Linked services, datasets and pipelines
- Copy Activity and data movement
- Parameters, variables and expressions
- Triggers and scheduling
- Incremental loading and metadata-driven pipelines

Hands-on: Build an automated ingestion pipeline using ADF.

Session 4 — Azure Databricks & Spark

- Azure Databricks workspace and compute
- Notebooks and cluster concepts
- PySpark data processing
- Delta Lake and Lakehouse architecture
- Integration with Azure Data Lake Storage
- Jobs and workflow orchestration

Hands-on: Build a PySpark pipeline in Azure Databricks.

Session 5 — Azure Synapse, Security & Monitoring

- Azure Synapse Analytics overview
- Serverless and dedicated SQL concepts
- Synapse pipelines and data integration
- Microsoft Entra ID fundamentals
- Azure Key Vault and secrets management
- Azure Monitor and Log Analytics

Hands-on: Secure and monitor a cloud data workflow.

Session 6 — Azure Production Data Engineering Capstone

- End-to-end Azure data architecture
- ADLS Gen2 → ADF → Databricks → Delta Lake/Synapse
- Batch and incremental data pipelines
- CI/CD and deployment concepts
- Security, monitoring and cost optimization

Capstone: Design and implement a production-style Azure data engineering solution.

Combined Learning Outcomes

- Develop practical Python skills for data engineering and automation.
- Build production-oriented pipelines using Azure services.
- Use Azure Data Factory, ADLS Gen2, Databricks and Synapse in an integrated architecture.
- Apply Python, PySpark and Delta Lake to real-world data processing.
- Implement security, monitoring, testing and deployment practices.
- Complete an end-to-end cloud data engineering capstone.

Recommended Training Structure: Python Fundamentals → Azure Fundamentals → Azure Data Engineering → Databricks & PySpark → Production Data Engineering Capstone.